

Theorems and Proofs

Previous Lecture

- *Equivalent predicates*
- *Equivalent quantified propositions*
- *Quantifiers and conjunction/disjunction*
- *Order of quantifiers*

A tale of two proofs

● "Proof" can mean different things:

- A formal, mechanical *derivation* in an *axiomatic* system, where each step has an explicit reason

Step	Reason
1. $\neg p$	premise
2. $p \vee q$	premise
3. q	Resolution (1,2)

- An informal written argument, often using implicit assumptions about mathematical structures, *which can in principle be formalized*

"The number x is not less than 0. Then x must be at least 0."

Axioms

● *Axiomatic systems combine three things:*

- *A collection of domains, predicates, and propositions*

e.g. domain: integers

$P(x,y)$: " $x > y$ "

$Q(x,y)$: " $x = y$ "

- *A set of obviously evident truths called the **axioms***

e.g. $\forall x. \forall y. (P(x,y) \vee P(y,x) \vee Q(x,y))$

- *A set of rules for making valid inferences*

e.g. modus ponens

$$p \rightarrow q$$

$$p$$

$$\therefore q$$

Zermelo-Fraenkel set theory

● *Domain: sets (next week!)*

• *Predicates:*

" $x = y$ "

" $x \in y$ "

• *Axioms*

Ext $\forall a \forall b [\forall x (x \in a \leftrightarrow x \in b) \rightarrow a = b]$

Pair $\forall a \forall b \exists c \forall x [x \in c \leftrightarrow (x = a \vee x = b)]$

Union $\forall a \exists b \forall x [x \in b \leftrightarrow \exists y (x \in y \wedge y \in a)]$

Pow $\forall a \exists b \forall x [x \in b \leftrightarrow \forall y (y \in x \rightarrow y \in a)]$

Inf $\exists a [\emptyset \in a \wedge \forall x (x \in a \rightarrow x \cup \{x\} \in a)]$

Sep _{φ} $\forall a \exists b \forall x [x \in b \leftrightarrow (x \in a \wedge \varphi(x, \vec{p}))]$

Repl _{φ} $\forall a [(\forall x \in a \exists! y \varphi(x, y, \vec{p})) \rightarrow \exists b \forall y (y \in b \leftrightarrow \exists x \in a \varphi(x, y, \vec{p}))]$

Found $\forall a [a \neq \emptyset \rightarrow \exists x \in a \neg \exists y (y \in x \wedge y \in a)]$

AC $\forall a [(\emptyset \notin a \wedge \forall x \neq y \in a \cap x \cap y = \emptyset) \rightarrow \exists c \forall x \in a \exists! y (y \in x \wedge y \in c)]$

Foundation of mathematics:
every theorem
can be* formalized using these
axioms & logical inferences

Modern foundations

- *ZFC Set theory is actually a pretty terrible foundation*
 - *Axiom of choice in particular is contentious*
- *Modern foundations based on (axiomatic) **type theory***
 - *Same meaning as "types" in C/Python*
 - *Curry-Howard correspondence: "programs are proofs"*
- *Theorem proving languages like **Lean, Rocq** use type theories in place of ZFC*
 - *Though ZFC can be defined within them*
- *For theoretical reasons, usually **intuitionistic/constructive***
 - *Can't assume law of the excluded middle!*

From formal to informal proofs

- *Axiomatic systems and formal proof are unwieldy for humans to write & understand*
 - *But good for computers / computer programs!*
- *Informal proofs are written in prose and*
 - *Don't explicitly state obvious inferences and axioms*
 - *Do explain assumptions, non-obvious axioms, or non-trivial inferences*

More formal

Less formal



premise	1. $\forall x (P(x) \rightarrow Q(x))$	<i>It is assumed that all men are mortal, and so if Socrates is a man, then Socrates is mortal.</i>	<i>Socrates is a man, and so Socrates is obviously mortal.</i>
instantiation	2. $P(\text{Socrates}) \rightarrow Q(\text{Socrates})$		
premise	3. $P(\text{Socrates})$	<i>Since Socrates is in fact a man, by the assumption Socrates must be mortal</i>	
modus ponens	4. $Q(\text{Socrates})$		

Mathematical taxonomy

- *Result: a proven mathematical fact*
- *Theorem: result of “great” importance*
e.g. “God exists”
- *Proposition: result of “lesser” importance*
e.g. “People exist”
- *Lemma: intermediate result used to prove other results*
e.g. “There exists an ordinal which is greater than all numbers”
- *Corollary: obvious consequence of another result*
e.g. “Not all ordinals are numbers”
- *Conjecture: a theorem which has not yet been proven*
e.g. “There are infinite parallel universes”

Proving theorems (or results)

- *Key: should be evidently formalizable!*
- *What this means in practice: should use known, valid rules of inference*

Theorem:

If $2x - 6 = 0$ then $x = 3$.

Logical form:

$\forall x. (2x - 6 = 0) \rightarrow (x = 3)$

Proof:

Take any number c such that $2c - 6 = 0$.

Then $2c = 6$, and hence $c = 3$.

As c is an arbitrary number this proves the theorem.



Universal generalization

Q.E.D

Formalization

- *Statement:* “If $2x - 6 = 0$ then $x = 3$.”
- *Predicates:* $P(x)$ - “ $2x - 6 = 0$ ”, $Q(x)$ - “ $2x = 6$ ”, $R(x)$ - “ $x = 3$ ”
- *Need to prove:* $\forall x (P(x) \rightarrow R(x))$
- *Axioms:* $\forall x (P(x) \rightarrow Q(x)), \quad \forall x (Q(x) \rightarrow R(x))$

<i>Step</i>	<i>Reason</i>
1. $P(c)$	<i>assumption</i>
1. $\forall x (P(x) \rightarrow Q(x)), \quad \forall x (Q(x) \rightarrow R(x))$	<i>premises</i>
2. $P(c) \rightarrow Q(c), \quad Q(c) \rightarrow R(c),$	<i>univ. instantiation</i>
3. $R(c)$	<i>Modus Ponens</i>
4. $\forall x (P(x) \rightarrow R(x))$	<i>univ. generalization</i>

Direct Proofs

- *Direct proofs are used when we need to proof propositions like*

$$\forall x (P(x) \rightarrow Q(x))$$

- *Goal: prove that $P(a) \rightarrow Q(a)$ is a tautology for an **arbitrary** a .*

- *Proof structure:*

1. *Assume that $P(a)$ is true*
2. *Using axioms, previous theorems etc. prove that $Q(a)$ is true*
3. *Conclude that $P(a) \rightarrow Q(a)$ is true*
4. *Use the rule of universal generalization to infer*

$$\forall x (P(x) \rightarrow Q(x))$$

Example

$$\text{Even}(n) \leftrightarrow \exists k. n = 2k$$

$$\text{Odd}(n) \leftrightarrow \exists k. n = 2k+1$$

● Definitions:

n is **even** if and only if there exists k such that $n = 2k$

n is **odd** if and only if exists k such that $n = 2k+1$

● Theorem:

If n is odd, then n^2 is odd

● Proof:

- Let n be an arbitrary odd integer.

Exist. instantiation

- Then there exists k such that $n = 2k+1$.

Implicit axioms

- Now, $n^2 = (2k+1)(2k+1) = 4k^2 + 4k + 1 = 2(2k^2 + k) + 1$.

Exist. generalization

- Hence there exists k' such that $n^2 = 2k' + 1$.

- Therefore n^2 is odd

Q.E.D.

Proof by contrapositive

- Sometimes a direct proof of $\forall x (P(x) \rightarrow Q(x))$ doesn't work out
- In these cases, it can be easier to instead prove the equivalent *contrapositive*, $\forall x (\neg Q(x) \rightarrow \neg P(x))$

● Theorem:

If $3n + 2$ is odd, then n is odd

● Proof:

- *Let n be an arbitrary integer and assume n is not odd.*
- *Then there exists k such that $n = 2k$.*
- *Now, $3n+2 = 3(2k) + 2 = 6k + 2 = 2(6k + 1)$.*
- *Hence there exists k' such that $3n+2 = 2k'$.*
- *Therefore $3n+2$ is not odd*

Q.E.D.

Proof by Contraposition (cont)

- *Used when need to prove $\forall x (P(x) \rightarrow Q(x))$*
- *Goal: prove that $\neg Q(a) \rightarrow \neg P(a)$ for an arbitrary a*
- *Proof structure:*
 1. *Assume that $\neg Q(a)$ is true*
 2. *Using axioms, previous theorems etc. prove that $\neg P(a)$ is true*
 3. *Conclude that $\neg Q(a) \rightarrow \neg P(a)$ is true*
 4. *Conclude that $P(a) \rightarrow Q(a)$ is true by logical equivalence*
 5. *Use the rule of universal generalization to infer*
$$\forall x (P(x) \rightarrow Q(x))$$

Example

● *Theorem:*

If $n=ab$, then $a \leq \sqrt{n} \vee b \leq \sqrt{n}$

● *Proof:*

- *Let a and b be arbitrary*
- *Suppose $a > \sqrt{n}$ and $b > \sqrt{n}$*
- *Then $ab > \sqrt{n}\sqrt{n} > n$*
- *Thus n is not equal to ab*

Q.E.D.

Proof by Contradiction

- *Proofs by contradiction use the Rule of Contradiction*

$$\frac{\neg p \rightarrow F}{\therefore p}$$

- *Can be used to prove propositions of any form*
- *Proof structure:*
 1. *Assume $\neg p$.*
 2. *Infer a contradiction, i.e. F*
 3. *Conclude p .*
- *To infer F , usually infer both r and $\neg r$ for some proposition r*

Example

- **Definition:** *a barber is called **strict** if he shaves those and only those who do not shave themselves.*
- **Theorem.** *There is no strict barber.*
- **Proof.**
- *Assume the contrary: a strict barber (c such that $S(c)$) exists*
- *Does he shave himself – say $Q(c)$?*
- *If no ($\neg Q(c)$), then by definition he must shave himself ($Q(c)$)*
- *If yes ($Q(c)$), then by definition he must not ($\neg Q(c)$)*
- *Either way we have $Q(c) \wedge \neg Q(c)$, a contradiction*
- *We conclude that a strict barber does not exist*

Another Example

● *Definition:* a real number is said to be **rational** if it can be represented as a fraction $\frac{a}{b}$ where a, b are integers

● *Prove that $\sqrt{2}$ is irrational*

● *Proof*

Suppose that $\sqrt{2}$ is rational

Then there are integers a, b such that $\sqrt{2} = \frac{a}{b}$.

We may assume that a, b have no common divisor.

Squaring we obtain $a^2 = 2b^2$.

Since a^2 is even, a is also even, hence $a = 2c$ for some c .

Therefore $2b^2 = 4c^2$, and so $b^2 = 2c^2$.

Hence b is even.

We get that a and b have a common factor – 2. A contradiction.

Practice

Exercises from the Book:

7th edition: 13, 15, 23, 27 (page 79 – 80)

8th edition: 13, 15, 23, 27 (page 83 – 84)